AI Driven Edge Device for Screening Skin Lesion and Its Severity in Peripheral Communities

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***Abstract*—It is vital to treat any skin disorder as early as possible. Neglecting the rudimentary skin disease may lead to acute skin cancer. Skin diseases are mostly neglected in peripheral regions because of a lack of awareness and accessibility to dermatologists. Therefore, it is important to diagnose skin diseases promptly and take countermeasures to treat them effectively. The paper presents a novel method to detect skin disease and its severity using a mobile application. A dataset consisting of four classes - Eczema, Measles, Leprosy, and Healthy Normal Skin is chosen in this proposed work. The images are passed through two modified CNN architectures. The first layer is a modified Mobile Net V2 architecture that aids in predicting the type of skin disease, which is referred to as SkinLesion Net. The next layer that predicts the severity of the disease is named SeverityNet. Observant comparison on four different CNN architectures - VGG16,Inception V3, Xception and the proposed SkinLesion Net, is performed using this image dataset. Skin Lesion Net outperforms the other networks with 94.32% accuracy, 93.02% F1-Score, 93.53% Precision and 92.76% Recall. The model is lightweight, about 14 MB in size, which is appropriate for deployment into an Android application.**

***Index Terms*—Convolutional Neural Networks, Skin Lesion Classification, Mobile Net V2, AI driven edge based device**

I. Introduction

Skin is the largest organ of the body. Regional and environmental factors predominantly impact skin damage. Skin diseases linger over a long period, spreading gradually all over the skin. According to a recent survey in central India, the prevalence of skin diseases is 60% which is highly alarming [1]. Measles is a contagious disease caused by Morbillivirus. India's Measles cases are estimated to be over 36% of the world's, which is estimated to be a significant proportion [2]. Measles is diagnosed from serum samples and throat swab [3]. Eczema, also referred to as atopic dermatitis, is a skin condition due to which at least one in five children suffer in India [4]. It deteriorates the skin barrier, subsequently permitting the germs to settle in. Leprosy is a highly contagious disease induced by Mycobacterium leprae. Timely treatment will prevent physical disabilities and help the patient escape the social stigma instilled by most skin diseases. India continues to account for 60% of new leprosy cases reported annually, consistent with the WHO report in 2016 [5]. Dermatological complications are particularly prevalent in India's peripheral communities. Unfortunately, there are few resources and trained personnel in the peripheral regions to handle dermatological complications immediately. Consequently, conventional ways of diagnosing and treating skin lesions are time-consuming. As part of the research, we propose an automated dermatological image classification system that can classify three prevalent skin diseases in India according to severity - Measles, Leprosy, and Eczema.

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The data set is tested using four different fine-tuned state-of-the-art pre-trained architectures. - Mobile Net V2, VGG16, Inception V3, and Xception models. We utilize the pre-trained CNN-based classification algorithms by transferring learning and fine-tuning a few layers to extract suitable features from the images, achieving preferable accuracy. Among the tested models, SkinLesion Net (Fine-tuned Mobile Net V2 model) is the preferred network for the final deployment since it achieves the best accuracy of 94% for disease prediction and the SeverityNet (Fine-tuned Mobile Net V2 model) with 91% accuracy for severity prediction. Furthermore, the potent lightweight architecture of this model reduces the load on the mobile device occupying lesser space. This model is deployed into the Android Mobile Application. This is an additional advantage that features simple, easy-to-use widgets. This application provides a preliminary recommendation and notifies the user about the infection and its severity.

The major contributions of this present work include:

- This paper classifies the severity of a disease using a deep learning approach rather than using the image segmentation technique.
- The image segmentation technique gathers conclusions from the pixel properties. In contrast, the CNN model extracts features from an image, compares them with the existing image dataset, and provides a reliable solution.
- The pre-trained CNN architectures are fine-tuned to suit the dataset, and relevant features are extracted.
- As the proposed work is edge device-based, it can be accessed globally with an Android Smart Phone.

The rest of the paper is divided into different sections. Section II compares the works of the various authors and technical information about the technique used in the proposed work. Section III demonstrates the proposed model and the deployment into an android device.

Section IV analyses experimental analysis, results, and discussion, while finally, Section V has the conclusions.

II. Literature Survey and Background Work

Several authors used CNN-based models to classify a variety of skin diseases. Esteva et al. [6] have successfully implemented skin cancer classification using VGG 16 and VGG 19 models with 129,450 clinical images inclusive of 3,374 dermoscopy images. The authors classified melanoma

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and keratinocyte carcinoma using dermoscopy. They suggest broadening the scope of decision-making for dermatological specialists with smartphone inclusion to make it accessible to the end-user. R. Amelard et al. [7] proposed HLIFs modeled using the ABCD criteria used by dermatological specialists where each HLIF represents a human-observable characteristic. They used 206 images of skin lesions under unconstrained environmental conditions. HLIFs were modeled to describe some human-observable characteristic features.

N. Tajbakhsh et al. [8] concluded that the fine-tuned pre-trained models perform better than those built from scratch in terms of computational complexity. They prove their point by detecting polyps for early cancer diagnosis and achieving better accuracy than the built model. B. Ahmad et al. [9] explored a framework based on fine-tuning layers of ResNet152 and InceptionResNet-V2 models with a triplet loss rather than a traditional loss function. They classified four skin diseases (acne, dark circles, blackheads, and spots) with 14000 (including data augmentation) input images to train the model.

Li-sheng Weith et al. [10] employed pre-processing the images to remove noise and background disturbance. Grey-level co-occurrence matrix (GLCM) was formulated to segment images of skin diseases to separate texture and color characteristics on three skin diseases - dermatitis, herpes, and psoriasis. Finally, the support vector machine (SVM) method was used to classify the images.

The overall issues with existing works are summarized as:

- The previous works do not dwell on developing an edge device-based solution.
- They have concentrated more on the technique used to classify the disease rather than grading the severity of the detected disease.
- The existing work emphasises the need for a Mobile Application and the severity determination to give the user a complete skin condition diagnosis.

The comparison of the proposed work with the existing approaches is tabulated in Table I. All the current works classify a set of varied skin diseases using various CNN architectures. They also focus on image segmentation techniques to segregate the texture of the image. A combination of the CNN layers was utilized to build many state-of-the-art models. These models were pre-trained using images from ImageNet. This proposed work explores four CNN architectures that were pre-trained by transfer learning and fine-tuning a few layers.

1) **Mobile Net V2:** The salient features of Mobile Net V2 architecture are the inverted residual, depth-wise separable convolution and its light weight architecture [11].
2) **VGG 16:** The architecture is simple because the hyperparameters are fewer than the other pre-trained architectures [11].
3) **Inception V3:** Inception V3 model maps cross-channel correlations, and spatial correlations to multiple kernels instead of a single kernel [11].
4) **Xception:** The modified Inception model, the Xception model, has a pointwise Convolution layer followed by a depthwise convolution layer [11].
5) **Transfer Learning:** Transfer learning is a technique that minimizes the training complexity and time since the model is trained for a similar task [12].
6) **Fine Tuning:** Fine-tuning helps in adjusting the hyper-parameters according to the dataset available [12].

III. Proposed Methodology

The proposed work can be divided into four sections. The first section deals with pre-processing data, followed by constructing a Neural Network to classify the disease. The following section relates to the construction of a Neural Network to grade the severity. The performance metrics of the CNN models are also discussed. Finally the deployment of the Neural Network within an Android application is discussed. This is explained diagrammatically in Fig. 1.

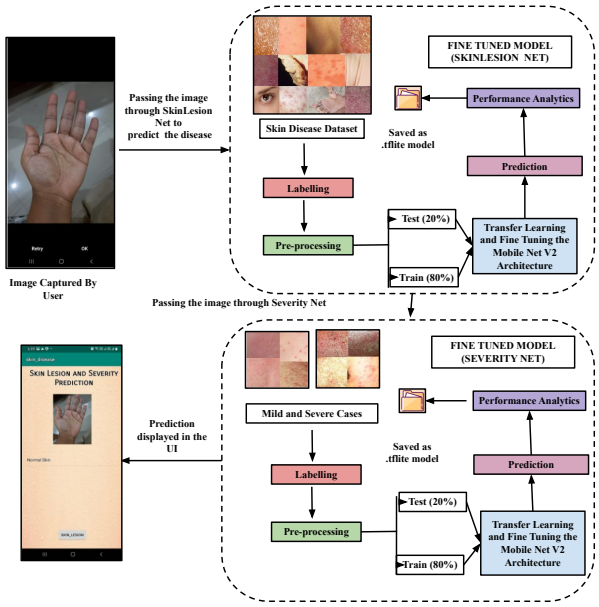


Fig. 1. The System Architecture

A. Pre-processing

The dataset is distributed with 80% of images devoted to training and 20% of images to the testing dataset. Based on this distribution, 1220 images are used for training and 304 images for testing. All the images are resized to $224 \times 224 \times 3$ (default dimensions for pre-trained models) to standardize the images. The images are further up-sampled using various data augmentation techniques. Up-sampling the dataset boosts the accuracy and robustness of the model. The dataset is augmented and up-sampled using the following parameters: 1) Re-scaling to normalize the image, 2) Zoom In and Zoom Out, 3) Width Shift, 4) Height Shift, 5) Rotation angle of 45°. The number of additional images added is the ratio of the number of training images to batch size. In the proposed work, approximately 47-48 new images are added per epoch, thus up-sampling the dataset for training.

Once the image is pre-processed the features are extracted using transfer learning and fine-tunig the existing CNN architectures which is discussed in the next section.

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TABLE I
COMPARISON OF THE EXISTING WORKS

Authors	Images	Diseases Classified	Edge Device	Model	Accuracy
Esteva et al.	129,450	Skin Cancer	No	VGG16 /VGG19	72.1%
R.Amelard et al.	206	Skin lesions	No	SVM Classifier	83.59%
N. Tajbakhsh et al.	1.2 million	Polyp Detection	No	AlexNet	-
B. Ahmad et al.	6144	Melonoma Detection	No	IncepResNet -V2	87.42%
Li-sheng Weith et al.	90	Skin Lesions	No	SVM Classifier	86.67%
Proposed Work	**1524**	**Skin Lesions and its Severity**	**Yes**	**SkinLesion Net and SeverityNet**	**94.32%**

B. Transfer Learning and Fine Tuning

The structural composition of the SkinLesion Net and SeverityNet is discussed in this section. The conventional Mobile Net V2 model has 1280 layers. This model is pre-trained to classify 1000 classes. Fig. 2 represents the fundamental bottleneck block of the Mobile Net V2 architecture with skip connections. The proposed work requires only four classifications. Thus, the training of the Mobile Net V2 model is preserved using the transfer learning technique, and only the final layers are fine-tuned. The architectural

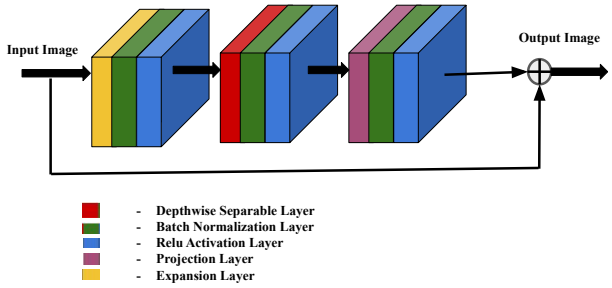


Fig. 2. The Inverted Residual Skip Connections of Mobile Net V2 Architecture

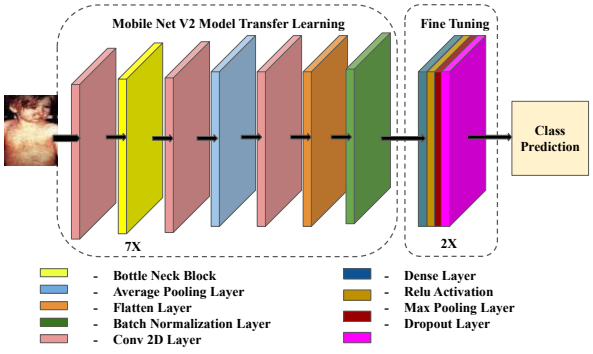


Fig. 3. The Architectural View of the Fine Tuned model

view of the SkinLesion Net is pictorially represented in Fig. 3. The existing architecture is fine-tuned in the following ways. The output from the pre-trained model is flattened as the first step. Subsequently, Batch Normalisation is applied to stabilize the training and achieve adequate accuracy within few epochs. A block of fully connected Dense Layer of size 512 and a Dropout Layer are concatenated. The next block comprises a fully connected Dense Layer of size 128 with L2 regularisation. Incorporating Dropout and L2 regularisation layers will bypass over-fitting on the training dataset. The final block is a fully connected Dense Layer of size four, with a softmax activation (2) [13].

$$\sigma(x_i) = \frac{exp(x_i)}{\sum_{n=j}^{K} exp(x_j)} \tag{1}$$

where x_i is the input feature and $\sigma(x_i)$ is the output feature. The ReLU activation function (3) [13] follows this Dense Layer.

$$ReLU(X) = max(0, X) \tag{2}$$

where X is the input feature and $RELU(X)$ is the output feature. Adamax optimiser is used to optimise the algorithm with a 0.0001 learning rate. This will ensure that model will take shorter steps while training. The learning rate is calculated in terms of log (4) [13].

$$LR_i = \sum_{j} \log(p_i, j) \tag{3}$$

The batch size is chosen as 32 after experimenting using different sizes. Categorical cross-entropy serves as the loss function. An illustration of the SkinLesion Net model utilized in the proposed work is shown in Fig. 3. This network classifies the type of skin lesion in an image. To determine the severity of the disease, SeverityNet is introduced in the final architecture. A dataset with fewer images is used to train this network. Since the disease is already classified in the SkinLesion Net, we utilize the transfer learning technique to preserve the features extracted. Only the final layers are fine-tuned to grade whether the disease is Severe or Mild. The calibration of SeverityNet is as follows. SeverityNet has an identical architecture as that of SkinLesion Net until the L2 regularization layer. A fully connected Dense layer of size two is appended to this. Due to the binary nature of the classification, binary cross-entropy is employed as the loss function along with sigmoid activation (4)

$$S(x) = \frac{1}{1 + \exp(x)} \tag{4}$$

where x is the input feature and $S(x)$ is the output feature. Similar procedure is applied on Mobile Net V2 model, VGG16 model, Inception V3 model, and Xception, and their performance is assessed to determine the best model to use for the mobile application. The two Fine-Tuned Mobile Net V2 models are chosen for the final deployment (SkinLesion Net and SeverityNet).

C. Performance Metrics

To evaluate the performance of both the architectures, Accuracy, Precision, Recall, F1-Score and the general Computational Complexity of the Mobile Net V2 architecture is

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utilized. The terms TP - denotes True Positives TN - denotes True Negatives FP - denotes False Positives FN - denotes False Negatives

1) Accuracy: Accuracy is the proportion of correctly classified labels with respect all the observations.

$$Accuracy = \frac{(TP + TN) \times 100}{(TP + FP + TN + FN)} \tag{5}$$

2) Precision: Precision is the fraction of true positive that the model classified as positive.

$$Precision = \frac{TP \times 100}{(TP + FP)} \tag{6}$$

3) Recall: Recall, also known as sensitivity, is the fraction samples classified as positive, among the total number of positive samples.

$$Recall(Sensitivity) = \frac{TP \times 100}{(TP + FN)} \tag{7}$$

4) F1-Score: F1 Score is the harmonic mean of Recall and Precision which measures the stability of the model

$$F1 - Score = \frac{2 \times (Recall \times Precision)}{(Recall + Precision)} \tag{8}$$

Here TP denotes the number of True Positive cases, TN is the number of True Negative cases, FP indicates the number of False Positive cases, and FN is the number of False Negative cases. The Computational costs of Mobile Net V2 architecture are:

$$C_{normal} = h_i.w_i.d_i.d_j.k \tag{9}$$

$$C_{separable} = h_i.w_i.d_i(d_j + k^2) \tag{10}$$

The C_{normal} denotes the cost of conventional convolution, $C_{separable}$ is the cost of depthwise separable convolution, i and j are the input and output layer index,h_i and w_i are the height and width of input feature maps d_i and d_j indicate the number of input and output feature map; finally, k is the filter size. The C_{normal} is quantified against $C_{separable}$ using the equation (11)

$$\frac{C_{normal}}{C_{separable}} = \frac{d_j k^2}{(d_j + k^2)}. \tag{11}$$

It is observed from Table II that the SkinLesion Net provides the best performance in terms of Accuracy, Precision, Recall, and F1-Score (6), (7), (8) and (9) [13]. The Accuracy and Loss Plots confirm the same. The equations (10), (11) and (12) [14] prove that the computational time complexity of the Mobile Net V2 architecture is considerable low. Thus SkinLesion Net, which is a modification to the same, suits the application. The Accuracy Plot of this SkinLesion Net took a big leap from the second epoch, as shown in Fig. 4, and the loss contrastingly took a steep downslope from the second epoch. The performance metrics validate the choice of using this model for this particular dataset. A similar analysis is done to find a suitable model to classify the severity of the disease. SeverityNet architecture is selected for the final severity prediction with 91% accuracy.

TABLE II
PERFORMANCE COMPARISON OF THE MODELS

Fine Tuned Model	Train Acc	Test Acc	Train Loss	Test Loss	F1-Score	Precision Score	Recall Score
VGG 16	82.35	81.58	0.5667	0.4679	88.98	97.62	82.23
Inception V3	85.70	91.12	1.2074	0.9682	93.26	94.58	92.43
Xception	90.60	93.42	0.2834	0.1797	92.71	93.70	92.10
SkinLesion Net	**83.96**	**94.32**	**1.1308**	**0.8601**	**93.02**	**93.53**	**92.76**

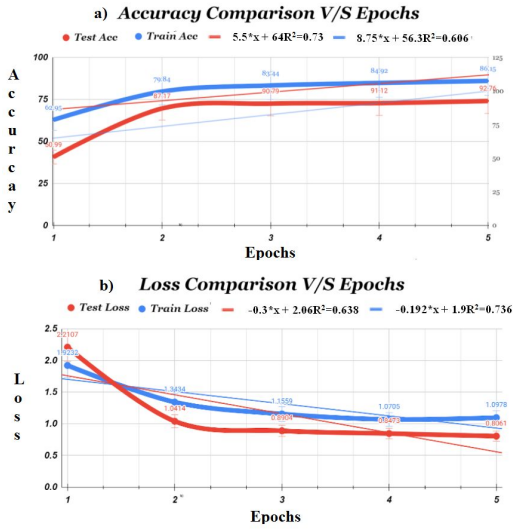


Fig. 4. The Plots of Accuracy and Loss

IV. Experiments and Results

A. Dataset

The dataset is collected from various open sources in the proposed work consisting of dermatological images certified by doctors. The sources include DermNet NZ, CNN-image-dataset, and other web scraped images [15]- [16]. Fig 5a) has the images of varying severity of Measles, Fig 5b) gives a sample image on Leprosy, while Fig 5c) illustrates the different grades of Eczema. These diseases are the opted diseases for classification. Healthy skin images are also included to detect an image without infection. This dataset consists of 1524 images belonging to 4 classes. These diseases are selected according to their prevalence in India and the availability of the dataset. The final data set is imbalanced with 159 images for Measles, 51 images for Leprosy, 1235 images for Eczema, and 81 images for Healthy Skin, tabulated in Table III. This imbalance in the data set is administered using image augmentation techniques. The images collected for the dataset are not identical in shape. Inherently the images are pre-processed before training.

TABLE III
DISTRIBUTION OF IMAGES IN THE DATA-SET

Disease	Number of Images	Train Images	Test Images
Measles	51	41	10
Eczema	1235	988	247
Leprosy	157	126	31
Normal Skin	81	65	16

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Fig. 5. a) Sample of Measles Disease, b) Sample of Leprosy Disease, c) Sample of Eczema Disease

B. Skin Lesion and Severity Detection

An image is passed through the CNN layers, each layer extracting certain features to classify the image. This is shown as a flow diagram in Fig. 6. The essential features are detected with each layer and learned by the model. Once the disease is classified using the SkinLesion Net, it is transferred to SeverityNet to classify its severity. Transfer

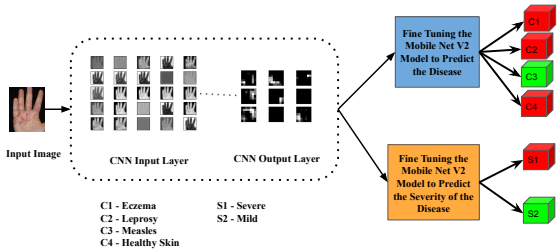


Fig. 6. Feature Extraction from an Image

learning and Fine-tuning techniques are applied to these models. Any other authors did not adopt this method to detect the severity of a disease to the best of our knowledge. The SkinLesion Net uses 1524 images while SeverityNet is trained on 45 images of Eczema and Measles each. The data augmentation and pre-processing are applied on the 90 images to up-sample the data. These deep learning models are finally incorporated into the Android Application using the TensorflowLite API.

C. Results and Discussions

In Fig. 7, the confusion matrices, describe the performance of the models for each disease.

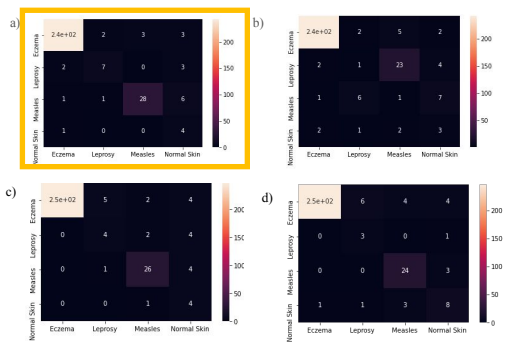


Fig. 7. Confusion Matrix for: a) Mobile Net V2 Model, b)VGG 16 Model, c)Xception Model and d)Inception Model

After analyzing the confusion matrices, it is evident that the fine-tuned SkinLesion Net performs the best with 1-3 misclassifications. Out of 304 test images, only 22 images are misclassified, achieving 94% accuracy. The only significant misclassification for this model is that six images of Measles are classified as Normal Skin. The modified VGG 16 accurately classifies only the images of Eczema. This model overfits the Eczema class since there are no misclassified images of Eczema. All the other classes have high misclassification rates. The Inception model works accurately for Eczema and Measles classes, while it has a high misclassification rate for Leprosy and Normal Skin. The modified Xception model is the next best model concerning this dataset. It has only 25 images misclassified. SkinLesion Net is preferred to the modified Xception model because of the lightweight architecture of the Mobile Net V2 occupying only 14MB size [17]. This model is converted into an API using the Tensorflow Lite library. The weights of the trained model are stored in the .tflite file. Since SkinLesion Net performed the best in classifying the disease, minor fine-tuning was employed to determine the severity of the disease, which lead to the construction of SeverityNet. Both the models are applied to the image one after another to determine the disease and its severity. The SeverityNet proved to achieve 91% accuracy to assess the severity of the predicted disease.

D. Android Mobile Application

The Mobile Application is developed using Android Studio. Deep Learning models are compatible with this application since it executes two models simultaneously with virtually no latency. The application was developed using Java on the backend. To diagnose the disease and its severity, separate image classifier classes were created in Java to integrate deep learning models. The models are saved in .tflite format and incorporated into the Mobile Application. Once an infection is predicted, the image is sent to the appropriate model with an if-else condition to detect severity. Regardless of the image's quality, the model can predict the disease based on an image of the affected part. The app requests permission to access the camera when it is installed for the first time. Fig. 8 presents screenshots of the developed prototype. Fig. 8a) is the initial blank UI with just the SKIN_LESION button. The application uses the phone's in-built camera to capture the skin image in response to being tapped, and disease is predicted. The image is passed to the SkinLesion Net to determine the skin lesion type, which is further passed through the SeverityNet to diagnose its severity. We can observe in Fig. 8b) and Fig. 8c) that Leprosy and Measles are detected, respectively, along with the seriousness. The immediate first aid solutions are also displayed to notify the user about preliminary aid. Because smartphones are commonly, a simple user interface, comprehensible diagnosis reports in an Android application that functions without latency make this application very user-friendly.

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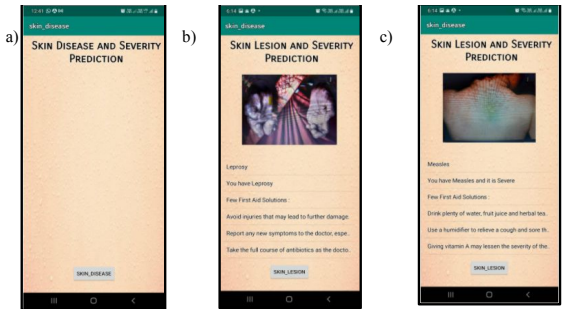


Fig. 8. Mobile Application Screenshots detecting Disease and the Severity

V. Conclusion

The proposed work aims at automating skin disease classification and severity prediction. This is accomplished using a CNN architecture in the backend coupled with a user-friendly Mobile Application. Eczema, Measles, Leprosy are the diseases considered for classification in this paper. We evaluate the performance of all fine-tuned models based on metrics like Accuracy, Loss, Precision, F1-Score, and Recall. The SkinLesion Net outperforms the rest of the models. This model achieves 94% accuracy. In addition to accurate skin disease classification, the SeverityNet architecture also predicts severity with 91% accuracy. As opposed to diagnosing skin disorders conventionally, this method saves time and money for the user. As this application evolves, we expect the scope to add the ability to diagnose other diseases and display a list of specialists for the diagnosed condition. All the data can be logged into a cloud server with the consent of the user, generalized to preserve their privacy, to further allow the researchers to analyze the pattern of skin diseases in a region. To summarize, developing similar applications and legalizing the same can create a breakthrough in the healthcare domain that can benefit humanity.

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